

DESCRIPTION AND DISCUSSION ON DCASE 2026 CHALLENGE TASK 2: NOISE-AWARE UNSUPERVISED ANOMALOUS SOUND DETECTION FOR MACHINE CONDITION MONITORING

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ABSTRACT

This paper presents an overview of DCASE 2026 Challenge Task 2, titled “Noise-aware unsupervised anomalous sound detection (UASD) for machine condition monitoring.” Reliable detection under noisy conditions is crucial for practical deployment, but previous DCASE Task 2 settings provided limited information about environmental noise, potentially limiting UASD performance in highly noisy situations. To address this limitation, DCASE 2026 allows participants to exploit two-channel audio samples simultaneously captured at locations near and far from the target machine. Since the distant microphone is expected to contain relatively stronger environmental noise and weaker direct machine sounds, it may help distinguish environmental noise components from the target machine sounds. We received 173 submissions from 52 teams, and an analysis of these submissions has been made in this paper. The results revealed that using the second channel can improve detection performance under noisy conditions. Specifically, the first ranked team used a self-supervised learning approach that tries to directly compute an SSL representation of the clean target machine sound from the two-channel recordings.

Index Terms— anomaly detection, acoustic condition monitoring, domain shift, first-shot problem, DCASE Challenge

1. INTRODUCTION

Anomalous sound detection (ASD) [1–4] involves determining whether the sound emitted from a target machine is normal or anomalous. This capability plays a crucial role in automating the detection of mechanical failures, which is essential in the era of the fourth industrial revolution and AI-driven factory automation.

One of the key challenges in developing ASD systems lies in the scarcity and limited diversity of anomalous samples available for training. To address this, the first ASD task was introduced in the DCASE Challenge 2020 Task 2 [5], focusing on “unsupervised ASD (UASD),” which aimed to detect unknown anomalous sounds using only normal sound samples for training. Building on this, subsequent challenges in 2021 and 2022 [6, 7] tackled the issue of domain shifts to enable broader application of ASD systems. Domain shifts refer to discrepancies between data from the source and target domains, arising due to variations in machine operational conditions or environmental noise. The following tasks from 2023 to 2025 (“first-shot” UASD) [8, 9] targeted a realistic setting where systems must detect anomalies for entirely novel machine types without access to similar-type data for training or hyperparameter tuning. This reflects rapid-deployment scenarios in which collect-

ing diverse training or test data, especially anomalous samples, is infeasible, and therefore manual test-driven tuning is unrealistic.

While various methods have been developed to improve UASD performance under such task settings, achieving high detection performance under noisy conditions remains an important challenge in real-world applications. To address this, the DCASE 2026 Challenge Task 2 focuses on noise-aware UASD, where two-channel audio samples captured near and far from the target machine are provided. The distant microphone can serve as a noise-reference signal since it generally contains less direct target machine sounds and relatively stronger environmental noise, which can potentially be leveraged to improve robustness against noise. In this paper, we provide explanations on this task and discuss the challenge results.

2. NOISE-AWARE UNSUPERVISED ANOMALOUS SOUND DETECTION UNDER DOMAIN SHIFT

Consider an audio clip $\mathbf{x}$, which contains sounds produced by a machine. The objective of the ASD task is to classify the machine as either normal or anomalous by calculating an anomaly score $\mathcal{A}_\theta(\mathbf{x})$ using an anomaly score calculator $\mathcal{A}$ with parameters θ . The input of $\mathcal{A}$ can be the audio clip $\mathbf{x}$ with or without additional information such as labels indicating the operation condition of the machine. The machine is then determined to be anomalous when $\mathcal{A}_\theta(\mathbf{x})$ exceeds a pre-defined threshold ϕ as

$$\text{Decision} = \begin{cases} \text{Anomaly} & (\mathcal{A}_\theta(\mathbf{x}) > \phi) \\ \text{Normal} & (\text{otherwise}). \end{cases} \quad (1)$$

The primary difficulty in this task is to train the anomaly score calculator with only normal sounds (UASD). The DCASE 2020 Challenge Task 2 [5] was designed to address this issue, and all the following tasks stand on this UASD setting. In addition to this basic UASD setting, this year’s challenge has three key features essential for the practical implementation of ASD systems. The first two features were introduced in the previous challenges, and the third feature is newly introduced in this year’s challenge.

The first feature is the domain-shift problem. Domain shifts are changes in conditions between training and testing that alter the distribution of the observed sound data. They can arise from differences in operating speed, machine load, heating temperature, microphone arrangement, environmental noise, and other factors. We define two domains: the **source domain**, the original condition with sufficient training data, and the **target domain**, another condition with only limited samples. This year’s task follows the 2022 to 2025 Task 2 [7–9] setting, where the domain information is assumed to be unknown in the test phase and anomalies from both

domains have to be detected with a single threshold. In this case, domain generalization is required to achieve good performance.

The second feature is addressing the “first-shot problem”. For the rapid development of ASD systems in real-world scenarios, solving ASD (a) against completely novel machine types (b) with only one set of training data (c) without handcrafted tunings that depend on test data, is highly important. This is because in real-world scenarios, customers may only possess a single novel machine, and collecting test data, especially anomalous samples, for handcrafted tuning may be infeasible. This problem setting is referred to as the “first-shot problem”, and the Task 2 2023 to 2025 [8–10] was organized based on this problem setting. This was implemented by introducing two features to the dataset: (i) The development dataset (which contains both training and test data with ground truth labels for the participants to develop their systems) and evaluation datasets (which do not contain ground truth labels, and are used for final evaluation of the systems) consist of entirely different sets of machine types, and (ii) Each machine type in the dataset contains only a single set of training and test data.

The third feature is the focus on noise robustness. Detecting anomalous machine sounds in noisy environments is important because real-world settings often contain various noise sources. Although previous DCASE Task 2 Challenges included noisy conditions, achieving high detection performance under such conditions remained challenging in some cases. To encourage the development of noise-robust UASD systems, this year’s task provides additional information related to noise. Specifically, while previous Task 2 datasets used single-channel audio, this year’s training and test samples are two-channel recordings captured simultaneously by microphones placed near to and far from the target machine. Participants may use either or both channels as input. Since the distant microphone is expected to contain less sound from the target machine and relatively stronger environmental noise, it may help identify which components of the recording correspond to environmental noise. Note that this setting is complementary to the DCASE 2025 setting. In DCASE 2025, supplemental data for improving noise robustness were provided as either clean target-machine recordings or noise-only recordings, which is feasible when the target machine or noise sources can be stopped in advance. In contrast, the DCASE 2026 setting applies when such recordings are unavailable but two microphones can be installed.

3. TASK SETUP

3.1. Dataset

The dataset for this task is divided into three categories: the **development dataset**, the **additional training dataset**, and the **evaluation dataset**. The development dataset contains seven machine types, while the additional training and evaluation datasets include five machine types, with each machine type consisting of a single section. A **machine type** refers to the category of machines, such as fans or gearboxes, and a **section** refers to a unit of data comprising a training set and its corresponding test set, for which performance is evaluated separately.

All recordings are two-channel, lasting 6 to 16 seconds, and have a sampling rate of 16 kHz. The machine sounds were recorded in a laboratory environment using two microphones placed at different distances from the target machine, with one microphone positioned close to the machine and the other farther away. Environmental noise was first recorded in factories or suburban areas, then played back through loudspeakers in the laboratory and rerecorded

using the same microphone setup as that used for the machine-sound recordings. The microphone setup, such as the distance between the microphones and the target machine, and laboratory arrangement differ across machine types, but the same setup was used consistently within each machine type.

The **development dataset** provides seven machine types (fan, gearbox (Emu), bearing (Emu), slide rail (Emu), valve (Emu), ToyCar, ToyCar (Emu)). Here, the suffix “Emu” indicates that the two-channel recordings were emulated rather than directly recorded. Each machine type has one section that contains a complete set of the training and test data. Each section contains (i) 990 normal clips from a source domain for training, (ii) 10 normal clips from a target domain for training, and (iii) 100 normal clips and 100 anomalous clips from both domains for the test. To assist participants, domain information (source/target) was included in the test data. For three machine types (ToyCar (Emu), fan, gearbox) details regarding the operational or environmental conditions were provided in the file names and attribute CSV files. For the remaining four machine types, these attributes were not disclosed. For five machine types (gearbox (Emu), bearing (Emu), slide rail (Emu), valve (Emu), ToyCar (Emu)), impulse responses were measured from the target machine position and from each loudspeaker position to the two microphones. The audio was then simulated by convolving these impulse responses with previously recorded machine sounds and environmental noise. All other data were recorded in real conditions using the setup described above.

The **additional training dataset** provides novel five machine types (ToyDrone, ToothBrush, SewingMachine, Sander, BlowerDustCollector). Each section consists of (i) 990 normal clips in a source domain for training, and (ii) 10 normal clips in a target domain for training. For three machine types (ToyDrone, Sander, BlowerDustCollector), attributes were provided in this dataset. For the other two machine types (ToothBrush, SewingMachine), attributes were concealed. For all machine types in this dataset, the audio clips were directly recorded in the laboratory as described above. The **evaluation dataset** provides the test clips that correspond to the additional training dataset, e.g. data of the same machine types as the additional training dataset. Each section consists of 200 test clips, none of which have ground-truth labels (i.e., normal or anomaly), domain information, or attribute information. Participants are required to train a model for each new machine type using only a single section per machine type.

3.2. Evaluation metrics

To assess overall detection performance, we employed the area under the receiver operating characteristic curve (AUC). Additionally, we used the partial AUC (pAUC) to evaluate performance in a low false-positive rate range $[0, p]$, where we set $p = 0.1$. Let m and n denote the index of a machine type and a section respectively, and $d \in \{\text{source}, \text{target}\}$ a domain. To evaluate each system under the domain generalization setting, we compute the AUC for each domain and pAUC for each section as

$$\text{AUC}_{m,n,d} = \frac{1}{N_d^- N_n^+} \sum_{i=1}^{N_d^-} \sum_{j=1}^{N_n^+} \mathcal{H}(\mathcal{A}_\theta(x_j^+) - \mathcal{A}_\theta(x_i^-)), \quad (2)$$

$$\text{pAUC}_{m,n} = \frac{1}{\lfloor p N_n^- \rfloor N_n^+} \sum_{i=1}^{\lfloor p N_n^- \rfloor} \sum_{j=1}^{N_n^+} \mathcal{H}(\mathcal{A}_\theta(x_j^+) - \mathcal{A}_\theta(x_i^-)), \quad (3)$$

where $\lfloor \cdot \rfloor$ is the flooring function and $\mathcal{H}(y)$ returns 1 when $y > 0$ and 0 otherwise. Here, $\{x_i^-\}_{i=1}^{N_d^-}$ are the normal test clips in domain d in section n of machine type m and $\{x_j^+\}_{j=1}^{N_n^+}$ are all the anomalous test clips in section n of machine type m . N_d^-, N_n^-, N_n^+ represent the number of normal test clips in domain d , normal test clips in section n , and anomalous test clips in section n , respectively.

The official score Ω is given by the harmonic mean of the AUC and pAUC scores overall machine types and sections:

$$\Omega = h \left\{ \text{AUC}_{m,n,d}, \text{pAUC}_{m,n} \mid m \in \mathcal{M}, n \in \mathcal{S}(m), d \in \{\text{source}, \text{target}\} \right\}, \quad (4)$$

where $h\{\cdot\}$ represents the harmonic mean, $\mathcal{M}$ is the set of given machine types, and $\mathcal{S}(m)$ represents the set of sections for machine type m . Since the dataset in 2024, $\mathcal{S}(m)$ contains only 00.

3.3. Baseline systems and results

The task organizers offer a baseline system using Autoencoders (AEs) with two operating modes, identical to the 2023 Task 2 baseline. While this year's data contains two-channel recordings, the baseline system uses only the first channel as input. Both modes use Autoencoders for training, but they differ in anomaly score computation. This paper presents the system and its detection performance; details can be found in [11].

3.3.1. Autoencoder training

The AE is trained for both operating modes using log-mel-spectrograms of training sound clips $X = [X_1, \dots, X_T]$, where $X_t \in \mathbb{R}^F$ for $t = 1, \dots, T$ represents frame-wise feature vectors at frame t , where $F = 128$ and T is the number of mel-filters and time-frames, respectively. For input, $P = 5$ consecutive frames are concatenated as $\psi_t = [X_t^T, \dots, X_{t+P-1}^T]^T \in \mathbb{R}^D$ for each t , with $D = P \times F = 640$. Model parameters are trained by minimizing the mean squared error (MSE) between the input ψ_t and the reconstructed output $r_\theta(\psi_t)$ for all inputs from the training data.

3.3.2. Simple Autoencoder mode

This mode uses the mean MSE of all features derived from the given sound clip as its anomaly score, e.g.,

$$A_\theta(X) = \frac{1}{DK} \sum_{k=1}^K \|\psi_k - r_\theta(\psi_k)\|_2^2, \quad (5)$$

where $K = T - P + 1$, and $\|\cdot\|_2$ represents ℓ_2 norm.

3.3.3. Selective Mahalanobis mode

In this mode, the anomaly score is defined based on the Mahalanobis distance between the input and reconstructed feature as

$$A_\theta(X) = \frac{1}{DK} \sum_{k=1}^K \min\{D_s(\psi_k, r_\theta(\psi_k)), D_t(\psi_k, r_\theta(\psi_k))\}, \quad (6)$$

$$D_s(\cdot) = \text{Mahalanobis}(\psi_k, r_\theta(\psi_k), \Sigma_s^{-1}), \quad (7)$$

$$D_t(\cdot) = \text{Mahalanobis}(\psi_k, r_\theta(\psi_k), \Sigma_t^{-1}), \quad (8)$$

where $\Sigma_s^{-1}, \Sigma_t^{-1}$ are the covariance matrices of $r_\theta(\psi_k) - \psi_k$ for the source and target domain data of each machine type, respectively.

Table 1: Baseline results for development dataset.

Machine type	Mode	AUC [%]		pAUC [%]
		Source	Target	
ToyCar (Emu)	MSE	69.62 ± 9.95	61.20 ± 6.34	55.89 ± 3.64
	Mahal.	69.49 ± 1.71	66.62 ± 6.74	53.47 ± 2.28
ToyCar	MSE	75.62 ± 1.98	37.87 ± 1.37	54.03 ± 0.70
	Mahal.	77.28 ± 1.57	53.17 ± 2.98	58.25 ± 0.53
bearing (Emu)	MSE	62.34 ± 1.09	59.56 ± 0.83	59.85 ± 0.23
	Mahal.	65.92 ± 1.74	62.28 ± 1.31	60.42 ± 0.37
fan	MSE	61.45 ± 0.66	46.94 ± 0.52	53.33 ± 0.37
	Mahal.	60.00 ± 4.09	45.09 ± 1.76	52.29 ± 0.32
gearbox (Emu)	MSE	68.23 ± 1.71	49.78 ± 0.75	52.94 ± 0.69
	Mahal.	74.48 ± 2.01	52.74 ± 1.90	53.97 ± 0.59
slider (Emu)	MSE	67.25 ± 1.11	45.05 ± 0.95	50.38 ± 0.40
	Mahal.	66.36 ± 0.45	49.18 ± 0.51	50.36 ± 0.17
valve (Emu)	MSE	67.74 ± 1.45	68.78 ± 1.03	55.08 ± 0.80
	Mahal.	56.60 ± 1.39	56.50 ± 1.49	50.20 ± 0.70

3.3.4. Results

Tables 1 present the AUC and pAUC results for the two baseline systems on the development dataset, with the averages and standard deviations computed from five independent trials.

4. CHALLENGE RESULTS

4.1. Overall results

We received 173 submissions from 52 teams. Among them, 28 teams outperformed both baseline systems, demonstrating that even beating the baseline is a challenging task. Figure 1 shows the AUC results for the top 10 teams in the evaluation dataset. Each team's performance varied across machine types. For example, team performance varied substantially for SewingMachine and ToothBrush. This shows that achieving stable performance across machine types was important for attaining a high ranking. Figure 2 shows the harmonic mean of AUC across the top 30 teams. The comparison between the development and evaluation datasets shows that achieving high performance on the development dataset does not necessarily guarantee high performance on the evaluation dataset, which reflects the fact that these datasets had different machine types. The comparison between the source and target domains shows a trade-off between the two domains. Nonetheless, the top-ranked team [12] achieved high performance on both domains.

4.2. Novel approaches in the top-ranked teams

This year, most of the top-ranked teams used audio encoders (either frozen pretrained models or models fine-tuned using auxiliary label classification tasks) followed by a lightweight anomaly scorer, such as k-NN distance, similar to the trend in previous years [10]. In this section, we discuss the novel approaches that were used on top of this common approach.

4.2.1. Approaches for exploiting two-channel audio

Among the top 28 teams that outperformed the baseline systems, 24 used the newly introduced second channel, recorded by a distant microphone. Their methods can be broadly grouped into three approaches. Among these, a particularly noteworthy approach was the second approach, adopted by the 1st-ranked team, in which a target-sound representation is directly learned from two-channel audio.

The first approach is a cascaded framework that uses the far-microphone signal as a noise reference [13–24]. In this approach, signal processing techniques are first applied either to suppress

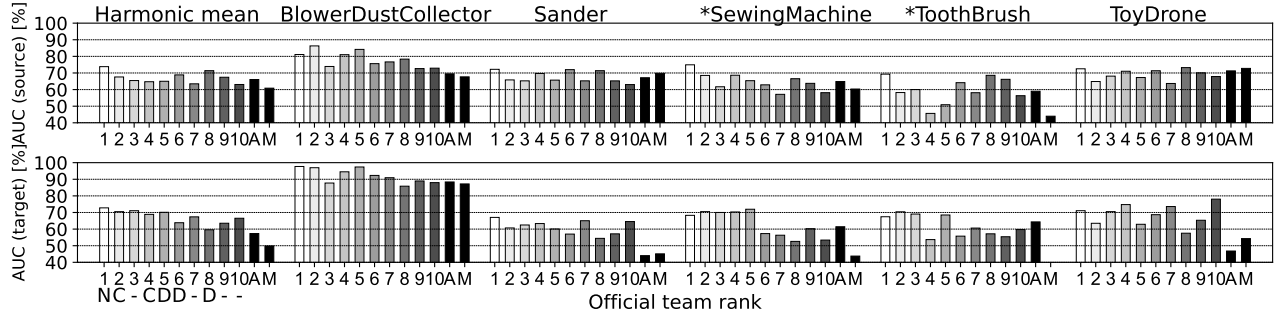


Figure 1: Evaluation results of top 10 teams in ranking. Average source (top) and target-domain AUC (bottom) for each machine type (“*” indicates that attributes are hidden). Labels “A” and “M” on denote simple Autoencoder and selective Mahalanobis mode, respectively. C, N, and D indicate the use of cascaded approaches, noise-aware SSL, and data augmentation, respectively.

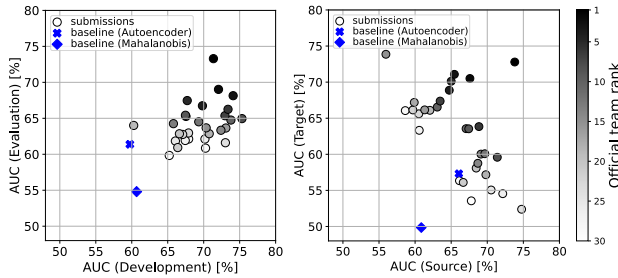


Figure 2: Comparison of harmonic mean AUC across teams (left: development vs. evaluation, right: source vs. target domain).

environmental noise in the near-microphone recording or to enhance the target machine sound. Then, the processed signal is fed into a UASD system. The methods used in the first stage varied across teams, including Kalman filtering [13] (2nd-ranked), spectral subtraction [14–16], Wiener filtering [17–20], MMSE-based filtering [21], and acoustic noise cancellation [22, 23]. By enhancing the target machine sound, this approach can emphasize differences between normal and anomalous machine sounds that could lead to performance improvements. A potential drawback, could be that these enhancement methods may introduce artifacts or leave residual noise in the processed signal. If the negative impact of such artifacts outweighs the benefit of emphasizing the target sound, the overall performance gain may be limited. This may explain why the performance of teams using this approach varied considerably, and careful design and application may therefore be important.

The second approach, adopted by the 1st-ranked team [12], was referred to as noise-aware audio self-supervised learning (SSL). In this method, an audio encoder takes two-channel audio as input and is trained on simulated data from external datasets to extract an SSL embedding of the clean target sound. Unlike the first approach, it does not use an explicit enhancement stage, but directly estimates the embedding of the target sound. This design may have contributed to its top-ranked performance by reducing the effect of environmental noise without introducing errors related to the enhancement processes. Its performance was also stable across machine types. One point that could be further considered is how much this method depends on the realism of the simulated recording conditions used for training. However, because the microphone placement relative to the target machine can often be controlled in practice, this dependency may partially be manageable.

The third approach used the second channel for data aug-

mentation [25–28]. In this setting, the far-microphone signal is treated as an additional training sample, and the encoder is trained with a channel-classification objective alongside the conventional machine-type or attribute classification objectives. Since the main difference between the two channels lies in how strongly the target machine sound is captured, this auxiliary task may help the encoder learn features related to the target sound. At the same time, this approach treats the two channels more independently and may not fully exploit their paired near-far relationship. Overall, many teams leveraged the second channel to improve robustness to noise, highlighting the potential of the two-channel setting.

4.2.2. Other techniques

We also highlight several techniques that were seen among the top ranking teams. First, several top-ranking teams used frequency-wise memory banks [29] for anomaly scorers [12, 30, 31]. Another commonly observed approach was to combine multiple anomaly score calculators applied to the audio embeddings [26, 27, 32].

5. CONCLUSION

We presented an overview of the DCASE 2026 Challenge Task 2. The task aims to develop ASD systems that work under noisy conditions, given two-channel audio recordings recorded at different distances from the target machine. We discussed new approaches that were used by the top-ranked teams, mainly focusing on how to exploit the two-channel recordings. Although we could not discuss every new approach, we hope that the technical reports will contribute to ongoing progress in anomalous sound detection.

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